

Relaxation time of a Markov Chain Analysis

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1 Abstract

We study the theory behind Markov chains and the conditions required to achieve a stationary distribution of such chains. We then look into the notion of relaxation time and observe the relationship between the two concepts via scientific implementations. Further, by exploring a variety of everyday applications, as well as focusing specifically on population growth, we highlight the abundance of Markov chains in real-life scenarios.

2 Introduction

Markov chains are an important mathematical tool in stochastic processes used to unravel predictions regarding the probability of an event occurring solely depending on the current state or value of a variate. Moving from one state in a Markov chain to the next is known as a transition. Each transition involves a probability and these transition probabilities can be listed in a transition matrix. Our first aim is to show that when certain conditions are met, the chain reaches stationary distribution, which can be found by utilising the said transition matrix.

Having looked into the concept of transition probabilities, we consider the notion of relaxation time. After experiencing a disturbance, a system requires a certain amount of time to return to the initial state. This time span depends on the transition probabilities and the number of states present in the system and is referred to as relaxation time. In this article, we explore how the concepts of Markov chains and relaxation time can be combined, by providing applications consisting of Spin systems, Cladograms, and Dyadic Tilings.

The final purpose of the article is to highlight the prevalence of Markov chains in real-world scenarios by exploring how widely the mathematical notion is utilised in various fields, such as biology, physics, medicine, and computer science. We illustrate several applications such as Page-Rank, Epidemics, HIV, and Diabetes, and primarily focus on the example of population growth, which estimates how the population transforms under certain events.

3 Theory on Markov chain

3.1 Markov chain and stationary distribution

Markov chains make predictions about future occurrences by only analysing the most recent event and not any past events. Under certain conditions, they are able to reach a steady state, known as the equilibrium, however, the said conditions are not always met. Utilizing mathematical processes helps determine whether the steady state has been reached.

3.1.1 Markov chain and transition matrix

A *Markov chain* is a stochastic model that depicts a sequence of possible events, where probabilities for the next state only depend on its previous event state, not the states before. Stated differently, it is a set of transitions and their probabilities, assuming no memory of past events.

Stochastic processes can be classified in different ways, for example, by their state space and index set.

Definition 3.1.1.1 A *state space* is a Euclidean space in which the variables on the axes are the state variables.

Definition 3.1.1.2 An *index set* is a set whose members label or index members of another set.

There exist two types of stochastic processes, discrete-time, and continuous-time. For discrete-time stochastic processes, the state space is the integers or natural numbers and the index set has a finite or countable number of elements, while, for continuous-time stochastic processes, the state space is a real line and the index set is some interval of the real line.

Markov chains and Markov processes are important classes of stochastic processes. A Markov chain is a discrete-time process, meaning changes to the system can only happen at discrete time values. An example of a Markov chain is the board game "Snakes and Ladders" in which changes in the game (the system) can only occur on someone's turn. A Markov process is the continuous-time version of a Markov chain. An example is the number of cars that have visited a drive-through at a local fast-food restaurant during the day since a car can arrive at any time rather than at discrete time intervals. Discrete-time stochastic processes, and therefore Markov chains, are considered easier to study because continuous-time processes require more advanced mathematical techniques, particularly due to the index set being uncountable.

Consider a stochastic process $\{X_t, t = 0, 1, 2, \dots\}$ that takes a finite number of possible values. Write $I = \{i, j, k, \dots\}$ for a finite state space. If $X_t = i$, then the process is said to be in state i at time t . We suppose that whenever the process is in state i , there is a fixed transition probability p_{ij} that it will next be in state j . That is, we suppose that

$$P\{X_{t+1} = j | X_t = i, X_{t-1} = i_{t-1}, \dots, X_1 = i_1, X_0 = i_0\} = p_{ij} \quad (1)$$

for all states $i_0, i_1, \dots, i_{t-1}, i, j$ and all $t \geq 0$. This is a Markov chain. Equation (1) can be interpreted as stating that the conditional distribution of any future state X_{t+1} , given past states $X_0, X_1, \dots, X_{t-1}$ and the present state X_t , is independent of the past states and depends only on the present state. This is called the *Markovian* property. The value p_{ij} represents the probability that the process will, when in state i , next make the transition into state j . Since probabilities are nonnegative and since the

process must make a transition into some state, we have that

$$P_{ij} \geq 0, \quad i, j \geq 0; \quad \sum_{j=0}^{\infty} P_{ij} = 1, \quad i = 0, 1, \dots$$

Let P denote the matrix of one-step transition probabilities p_{ij} , so that

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ \vdots & \vdots & \vdots & \\ P_{i0} & P_{i1} & P_{i2} & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix}$$

t -step transition probabilities can be denoted as $P(X_t = j | X_0 = i) = p_{ij}^{(t)}$, where $\mathbf{P}^{(t)} = \mathbf{P}\mathbf{P}\dots\mathbf{P}$ is the t -fold matrix product. [21]

3.1.2 Communication, irreducibility, and aperiodicity

State j is said to be accessible from state i if for some $t \geq 0$, $P_{ij}^{(t)} > 0$. Two states i and j accessible to each other are said to *communicate*, and we write $i \leftrightarrow j$.

Proposition 3.1.2.1 Communication is an equivalence relation. That is

- (i) $i \leftrightarrow i$,
- (ii) if $i \leftrightarrow j$, then $j \leftrightarrow i$,
- (iii) if $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.

We say that the Markov chain is *irreducible* if all states communicate with each other, i.e., it is possible, with non-zero probability, to move from any state in the state space to any other state.

A Markov chain is said to be *recurrent* if, starting from any state in the space, the probability of eventually returning to that state is 1.

A recurrent state i is called *periodic* if there is some integer $d > 1$ such that a state i can be visited only at time $\{d, 2d, 3d, \dots\}$ steps, that is with a period of d . In other words, the state i is said to have period d if $P_{ii}^t = 0$ whenever t is not divisible by d , where d is the greatest integer with this property. If $P_{ii}^t = 0$ for all $t > 0$, then define the period of i to be infinite. If $d = 1$, we call the chain *aperiodic*. Note that periodic behaviour of a chain, i.e., when the chain can only return to a state at multiples of some integer larger than 1, complicates the study of the limiting behavior of the chain. [21] [33] [35]

3.1.3 Stationary distribution

Assume the chain is *irreducible*. A fundamental result ([23] Theorems 1.7.7 and 1.5.6) is that there exists a unique *stationary distribution* $\pi = (\pi_i : i \in I)$, i.e., a unique probability distribution satisfying the *balance equations*

$$\pi_j = \sum_i \pi_i p_{ij} \quad \text{for all } j \in I. \quad (2)$$

The point of stationarity is that, if the initial position X_0 of the chain is random with the stationary distribution π , then the position X_t at any subsequent non-random time t has the same distribution π , and the process $(X_t, t = 0, 1, 2, \dots)$ is then called the *stationary chain*. [22]

Example Consider a Markov chain with three states 1, 2, and 3, and the transition matrix:

$$P = \begin{bmatrix} 2/5 & 2/5 & 1/5 \\ 1/5 & 3/5 & 1/5 \\ 1/5 & 2/5 & 2/5 \end{bmatrix}$$

Remark: all rows add up to 1 and no zeros present i.e., possible to go from one state to another and back.

The stationary distribution (π_1, π_2, π_3) is given by

$$\pi P = \pi \quad (3)$$

So we have

$$\begin{bmatrix} \pi_1 & \pi_2 & \pi_3 \end{bmatrix} P = \begin{bmatrix} \pi_1 & \pi_2 & \pi_3 \end{bmatrix}$$

We now solve this,

$$(2/5)\pi_1 + (1/5)\pi_2 + (1/5)\pi_3 = \pi_1$$

$$(2/5)\pi_1 + (3/5)\pi_2 + (2/5)\pi_3 = \pi_2$$

$$(1/5)\pi_1 + (1/5)\pi_2 + (2/5)\pi_3 = \pi_3$$

We get that $\pi_1 = 1/4, \pi_2 = 2/4, \pi_3 = 1/4$. This is the stationary distribution for the given transition matrix and chain.

Remark: $\pi_1 + \pi_2 + \pi_3$ must equal to 1.

For this example, all of the transition probabilities are non-zero. Therefore, it is clearly possible to move from any state to any other state in one step, so the chain is irreducible.

The stationary distribution plays the main role in asymptotic results, as follows.

Definition 3.1.3.1 An *ergodic* system or process includes or impinges on all points in

a given space given sufficient time, and can be represented statistically by a reasonably large selection of points.

Theorem 3.1.3.1 (The ergodic theorem: [23] Theorem 1.10.2) *Let $N_i(t)$ be the number of visits to state i during times $0, 1, \dots, t - 1$. Then for any initial distribution*

$$t^{-1}N_i(t) \rightarrow \pi_i \text{ a.s., as } t \rightarrow \infty \quad (4)$$

Theorem 3.1.3.2 (The convergence theorem: [23] Theorem 1.8.3) *For any initial distribution,*

$$P(X_t = j) \rightarrow \pi_j; \text{ as } t \rightarrow \infty, \text{ for all } j \quad (5)$$

provided the chain is aperiodic.

Many general identities for Markov chains can be regarded as aspects of the ergodic principle "time averages equal space averages", and are used to derive expressions for concepts like mean hitting times.[22] [23]

3.1.4 Non-stationary Markov chain

A "non-stationary Markov chain" is a Markov chain in which the transition matrix is a function of time so that $P(X_{t+1} = j | X_t = i)$ may not be equal to $P(X_{r+1} = j | X_r = i)$ when $t \neq r$. More commonly, this is called non-homogeneous rather than non-stationary.

3.1.5 Continuous-time chains

The theory of continuous-time Markov chains closely parallels that of the discrete-time chains discussed above.

A continuous-time chain is specified by *transition rates* ($q(i, j) = q_{ij}, j \neq i$) which are required to be non-negative but have no constraints on the sums.

Given the transition rates, define

$$q_i := \sum_{j: j \neq i} q_{ij} \quad (6)$$

and extend (q_{ij}) to matrix $\mathbf{Q}$ by putting $q_{ii} = -q_i$. The chain $(X_t : 0 \leq t < \infty)$ has two equivalent descriptions.

1. Infinitesimal description. Given that $X_t = i$, the chance that $X_{t+dt} = j$ is $q_{ij}dt$ for each $j \neq i$.

2. Jump-and-hold description. Define a transition matrix $\mathbf{J}$ by $J_{ii} = 0$ and

$$J_{ij} := q_{ij}/q_i, \quad j \neq i \quad (7)$$

Then the continuous-time chain may be constructed by the two-step procedure:

- (i) Run a discrete-time chain X^J with transition matrix $\mathbf{J}$.
- (ii) Given the sequence of states $i_0, i_1, i_2, \dots$ visited by X^J , the durations spent in states i_m are independent exponential random variables with rates q_{im}

The discrete-time chain X^J is called the *jump chain* associated with X_t .

The results in the previous section go over to continuous-time chains via several modifications - see Norris [23] Chapters 2 and 3 for details.

Given a discrete-time chain X with some transition matrix $\mathbf{P}$, one can define the *continuised* chain $\tilde{X}$ to have transition rates $q_{ij} = p_{ij}, j \neq i$. Many quantities are unchanged by the passage from the discrete-time chain to the continuised chain. In particular, the stationary distribution π and mean hitting times $E_i T_A$ are unchanged. Therefore, results stated in continuous time can often be immediately applied in discrete time, and vice versa. [22] [23]

4 Theory on Relaxation Time

When a system transitions from one state to another, it invariably encounters disturbances before reverting back to its initial state. The time frame in which the system remains under the influence of the disturbance is precisely the time needed for the system to relax before performing its next action, this is referred to as relaxation time.

4.1 Definition of Relaxation Time

The relaxation time of a Markov chain refers to the average duration for which a system remains in a stable state after being disrupted. It represents how quickly the system achieves stability. To begin, a random process is discretised, and these discrete states are used to define the Markov states. These Markov states determine the probabilities of transitioning between states. The relaxation time can vary based on the transition probabilities and the number of states involved. It plays a significant role in the analysis of complex systems and finds applications in diverse fields like physics, pharmacy, chemistry, computer science, and economics. When considering eigenvalues (λ_1 and λ_2) of a transition probability matrix for the Markov chain, excluding the value of 1, the relaxation time is calculated by selecting the maximum eigenvalue. The calculation is expressed as:

$$r = -\frac{\Delta t}{\log |\lambda_2|}$$

where $\lambda_2 > \lambda_1$ are both the two largest eigenvalues of a probability matrix other than 1.

When considering a stationary matrix of a Markov chain, the main eigenvalue is 1, while the remaining eigenvalues are complex. However, if the chain is reversible, the

eigenvalues become real. In this case, the relaxation time of that specific chain can be defined as follows:

$$\text{Relaxation time} = \begin{cases} \text{Maximum of the real parts of eigenvalues} & , \text{if the eigenvalues are real} \\ \text{Maximum of the moduli of eigenvalues} & , \text{otherwise} \end{cases}$$

$$r = \frac{1}{1 - \lambda_2}$$

, where λ_2 is the second largest eigenvalue of the chain.

Another important thing Schweinsberg shows in his article [14] is that relaxation time can be bounded above.

Let $\{X_t\}_{t \geq 0}$ and $\{Y_t\}_{t \geq 0}$ and consider a subset U of the leaves $\{1, 2, \dots, n\}$: From [14] we observe that there exists a constant $C_1 < \infty$ such that the set $B = \{x \in T_n : \text{diam}(x) \leq C_1 n^{1/2}\}$ has $\pi(B) \geq 1/2$, where $\pi(\cdot)$ is the uniform distribution on T_n . Then, there exists a finite constant $M_1 < \infty$ such that:

$$\tau_{\text{rel}}(X) \leq M_1 n^{5/2}$$

, where $\text{diam}(x)$ is the maximum distance between any two elements or vertices within that set.

The proof of this is too long but is proven in the attached article.[14]

To conclude this sub-section we will observe another upper bound of relaxation time:

Let $B \subset T_n$. Suppose that for all $x \in T_n$ and $y \in B$, γ_{xy} is a path in G , possibly random, from x to y which has at most L edges. Then,

$$\tau_{\text{rel}}(X) \leq \frac{4L}{\pi(B)} Q(e) \sum_{x \in T_n} \sum_{y \in B} (\pi(x)\pi(y) \times \Pr(e \in \gamma_{xy}))$$

where e is the maximum edge in E . [14, 24]

This is a graph theory-based approach to relaxation time in terms of Markov Chains and a powerful source of information about this particular aspect of mathematics.[14] By stationary probabilities, their relaxation time is being estimated. Schweinsberg shows an explicit way of calculating Markov Chains as well as relaxation time. A proven theorem in the article states that relaxation is of $O(n^2)$ complexity.[14]

Different applications of Markov Chains can give different bounds of their relaxation times. So by generalizing the procedure of every mathematician, a pattern can be observed. That pattern is the first hint of every scientist trying to prove another bound of relaxation time and that pattern is mainly distributed in the above two cases.

Generally, lots of applications of Markov Probability are using different approaches

to construct any sort of bound to relaxation time. So, by identifying such bounds this can be used to bound it furthermore such that it can be used in calculations later on or even estimate the value of it as precisely as possible.

4.2 Relationship between Markov chain and relaxation time

To gain a comprehensive understanding of the relationship between relaxation time and Markov chains, we are going to present an order of applications, where their connection is highlighted.

4.2.1 Mixing time

A common link between Markov chains and relaxation time is the relaxation time formula

$$\tau^{rel} = 1/(1 - \lambda_{n,2})$$

where $(1 - \lambda_{n,1}) > \lambda_{n,2}$ and λ represents the largest eigenvalues of the transition matrix. Note that relaxation time is the time needed for a Markov chain to reach its stationary distribution.

In general, in order to provide a relation between the two aspects, mixing time comes to the equation where in most cases $\tau^{rel} = O(\tau)$ where τ is the mixing time and $O(\tau)$ represents the order of magnitude of the mixing time.

Definition 3.3.1.1 *Mixing time* is the time needed for a Markov chain to approximately reach its stationary distribution π .

4.2.2 Application: Cladograms

A cladogram is an unrooted tree with $n \geq 4$ leaves and with $n - 2$ unlabeled internal vertices of degree 3. Cladograms are of major interest in the field of biology since it gives an evolutionary relationship between n species. A Markov chain on cladograms occurs when uniformly removing a random leaf and re-attaching this leaf to a random edge. A transition matrix is formed to represent this Markov chain and is shown below:

$$P_n(t, t') = \begin{cases} \frac{1}{n(2n-5)} & \text{if } t' \sim t \\ \frac{n}{n(2n-5)} & \text{if } t' = t \end{cases}$$

Then, we notice the equation for relaxation time, given in sub-section 4.1, but with $\lambda_{n,1} = 1 \geq \lambda_{n,2}$ and another for mixing time that is outside the scope of our paper.

We can observe how the probability is distributed on a cladogram with the help of the figure below:

As stated above, cladograms are a useful tool in representing phylogenetic relationships i.e., to which species other species are closely related to. This is an intuitive

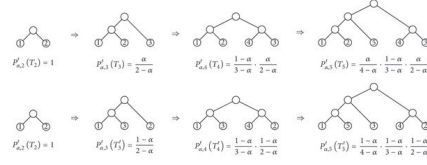


Figure 1: [28]

example of a Markov chain since the evolution of these species is affected only by the previous state and hence satisfying the definition of a Markov chain. What one can observe, is how each species is affected by another species through its evolution. All in all, this is a great representation of how a Markov chain functions in real life. This is illustrated by the figure below that shows possible phylogenetic relationships among hominoids - great apes:

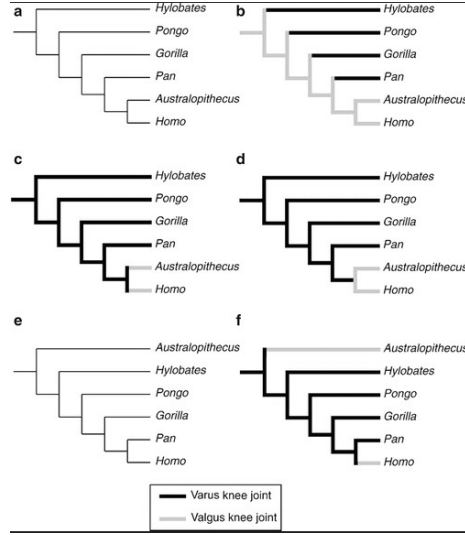


Figure 2: [29]

4.2.3 Application: Dyadic Tilings

A dyadic tiling of order n is a tiling of the unit square that yields in any rectangle through successive bisections.

The edge-flip Markov chain that represents such tilings is seen from [10]. In general, for this Markov chain, there is a strong relationship between relaxation time and mixing time, i.e. $\tau^{rel} \leq O(n^x)$ and $\tau \leq O(n^{x+1})$, with $x \in \mathbb{C}$, τ^{rel} is relaxation time

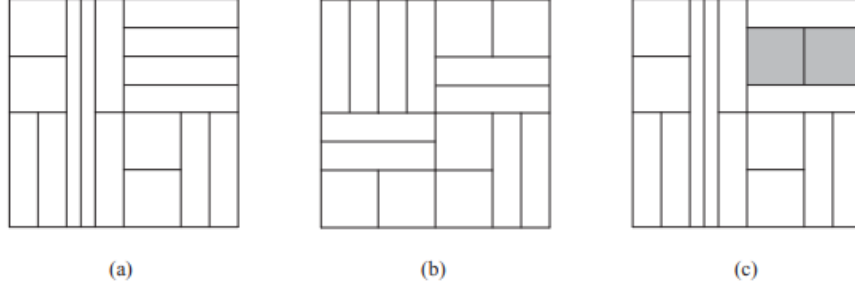


Figure 3: [10]

and τ is mixing time. Notice how relaxation time works as a bound for estimating mixing time throughout this application. Through section 2.2 of [10], we observe the time it takes for a Markov chain with its transition matrix to converge to its stationary distribution. This approach helped reach an approximation for mixing time that can only be calculated by bounding it with expressions for relaxation time. This result seen from [10] is as follows:

$$(\tau^{rel} - 1) \log\left(\frac{1}{2\epsilon}\right) \leq \tau(\epsilon) \leq \log\left(\frac{1}{\epsilon\pi_{min}}\right) \tau^{rel} \quad (8)$$

where $\epsilon \geq 0$ and $\pi_{min} = \min_{x \in \Omega} \pi(x)$. The transition probability, $T_n(p)$ for this particular application denotes the probability that there exists a set of available order n tiles that constitutes a tiling of the unit square. In order to compute the aforementioned probabilities we notice from [9], how each of vertical and horizontal tilings is available with a different probability for $n \in \mathbb{N}$. This constitutes another good application that links relaxation time and Markov chains since it shows clearly how mixing time is the most important link between them.

4.2.4 Mixing time revisited

In general, relaxation time is not something specific for all Markov chains since it is dependent on the transition probabilities of each Markov state and how it converges to its uniform stationary distribution. However, relaxation time is strongly related to mixing time. This is a helpful outcome because the mixing time of a Markov process can be difficult to estimate and relaxation time can help to provide us with bounds on mixing time and hence reach useful conclusions on Markov chains.

4.2.5 Application: Spin systems

We have seen how relaxation time is linked to Markov chains through mixing time. Now we provide an example of how to compute the relaxation time of a specific system in the field of physics and what that time represents.

”Markov random fields” are undirected graphs, graphs in which all edges are functioning in two directions (bidirectional), where the probability distribution of random variables is captured in each vertex and the edges represent the dependency relationship among the vertices, so the variables. Spin systems are an example of those models.

A spin system is a general framework for interacting spins, which can be thought of as little magnets pointing at each other (spins are not magnets in physics, but this comparison can make us understand their interpretation). It consists of a finite graph $G = (V, E)$ and a set S of spins, where the vertices are assigned elements from S . This assignment $W : V_G \rightarrow S$ is called configuration. The probability distribution of the set of all possible spin configurations, called Gibbs distribution, is given by $\mu(\sigma) = \frac{e^{-H(\sigma)}}{Z}$ [20], where Z is the partition function and H is the Hamiltonian, which depends on the spin values of each vertex and each pair of adjacent vertices. We will consider a spin system on an n -vertex d -dimensional cube of the integer lattice graph $\mathbb{Z}^d$, a graph where the vertices are arranged in a grid-like pattern and connected with lines that are not diagonal (figure 5).

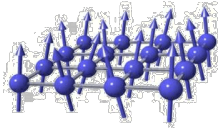


Figure 4: spin system [31]

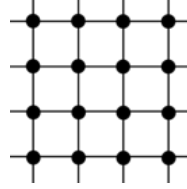


Figure 5: Lattice graph [32]

Among the most crucial models of spin systems, are the Swendsen-Wang (SW) dynamics which are a way to stimulate the behaviour of a magnetic system. The system is divided into clusters which were defined by Coniglio and Klein as a group of parallel neighbour spins that are connected by bonds[34]. Bonds are placed randomly with a probability of existence $p = 1 - e^{\frac{-2J}{(k_b T)}}$, where J is the coupling constant, it describes the exchange interaction energy between two neighbour spins in the cluster, k_b is the Boltzmann constant which has a defined value of 1.380649×10^{-23} Joule per Kelvin [37] and T is the critical temperature[19]. The system has q spin values. The clusters are either flipped up or down with equal probability and every spin has the same value as the cluster it belongs to. (SW) dynamics are non-local dynamics, this means that the configuration can be updated in only one step, so we do not need to go through each step one by one to make changes in the system.

We consider the easiest case where the dynamics are in one dimension. Let σ_i be the spins at a one-dimensional lattice graph $\mathbb{Z}$, where $i = 1, 2, \dots, L, L + 1$. Define the link variable $b_i = 1 - \delta_{\sigma_i, \sigma_{i+1}}$ which represents the probability of the existence of a bond between the adjacent spins σ_i and σ_{i+1} and δ is the Kronecker delta function which takes values +1 if σ_i and σ_{i+1} are the same spin and 0 otherwise. Let the bond

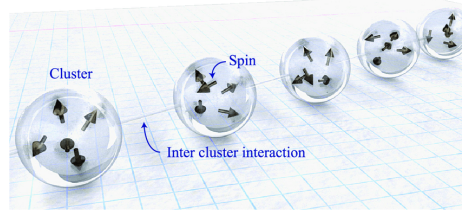


Figure 6: representation of spin clusters [30]

variables $n_i = 0, 1$ represent the absence or existence of a bond between adjacent spins. (Note: the Kronecker delta function is a function whereas the bond variables are constants). The joint probability distribution of spins and the existence of bonds is given by

$$\mathbb{P}(\sigma, n) = \prod_{i=1}^L [p\delta_{\sigma_i, \sigma_{i+1}}\delta_{n_i, 1} + (1-p)\delta_{n_i, 0}], \text{ where } p = 1 - e^{-\frac{2J}{k_b T}} \text{ as defined earlier.}$$

We apply 2 conditional probabilities:

1. Sample bonds given the spins: $\mathbb{P}(n | \sigma) = \frac{\mathbb{P}(\sigma, n)}{\mathbb{P}(\sigma)}$
2. Sample spins given the bonds: $\mathbb{P}(\sigma | n) = \frac{\mathbb{P}(\sigma, n)}{\mathbb{P}(n)}$

We will use the link variables instead of the spin variables, so

1. $\mathbb{P}(n | b) = \frac{\mathbb{P}(b, n)}{\mathbb{P}(b)}$
2. $\mathbb{P}(b | n) = \frac{\mathbb{P}(b, n)}{\mathbb{P}(n)}$

From research done by Jianqing Du, Bo Zheng, and Jian-Sheng Wang [19], we have that the conditional probabilities with respect to the link variables are the products:

$$\mathbb{P}(n | b) = \prod_i f(b_i, n_i) \quad \text{and} \quad \mathbb{P}(b | n) = \prod_i g(n_i, b_i)$$

where

$$f = \begin{pmatrix} 1-p & p \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad g = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{pmatrix}$$

The transition probability from one link configuration to another is then given by

$$W(b \rightarrow b') = \sum_n \mathbb{P}(b' | n) \mathbb{P}(n | b) = \prod_i w_{b_i, b'_i},$$

where $w \in W$.

The transition matrix is:

$$w = \begin{pmatrix} \frac{1+p}{2} & \frac{1-p}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

(The proof is outside of the scope of this article, for further research turn to the reference [19])

The eigenvalues of this matrix are 1 and $\frac{p}{2}$. We can extend the matrix to the whole configuration W, by taking the direct product of the transition matrix w for each site of the lattice, we get that each eigenvalue is given by $\lambda_m = \left(\frac{p}{2}\right)^m$, this requires knowledge about “ ϕ^4 theory” which is out of the scope of this paper, details can be found [18]. Notice that the first eigenvalue is always 1 and the second $\frac{p}{2}$, so the largest eigenvalue other than 1 is $\frac{p}{2}$. By the equations mentioned earlier in this article, we know that the relaxation time is given by $r = \frac{\Delta t}{\log |\lambda_2|}$ so the relaxation time, in this case, is $r = \frac{\Delta t}{\log |\frac{p}{2}|}$ where Δt is the time taken for the sampling process of bonds and spins. This gives precisely the time needed for the system to relax back to equilibrium before it transfers energy to its neighbour in the lattice.

5 Markov chain in real life

Markov chains have been noticed and researched in a variety of different fields, as already mentioned above. In order to highlight the importance of a Markov chain in the mathematical as well as the real world, we introduce some of its applications in real-life scenarios, which illustrate how Markov chains have helped people and circumstances on various occasions.

5.1 PageRank

PageRank (PR) is an algorithm used by Google Search to rank web pages in their search engine results. It provides a hierarchic order for the World Wide Web. The PageRank of a webpage is defined by a Markov chain and is the limiting probability that an infinitely dedicated random surfer visits any particular page. A surfer’s web link transition on a particular website can be modeled using first- or second-order Markov models and can be used to make predictions regarding future navigations as well as to personalize the webpages for an individual surfer.

The Web and its hyperlinks form a directed graph. Each link directs to another web, and each node is that webpage itself. The PageRank algorithm takes a random walk along the graph simulating a surfer going from one page to another by randomly choosing an outgoing link from one page to the next one at any point in time. Because the Web is not a strongly connected graph this process sometimes leads to dead ends, i.e., a surfer can get stuck on a page if there are no outgoing links or, if some webpage has no incoming links, the surfer can never visit that page. Utilizing a random walk, known as a Markov Chain, allows us to model the graph as a strongly connected graph.

During the walk, data such as how many times the surfer reaches a certain webpage over the entire operation is collected, and a ranking for each webpage, whereby a higher ranking means that the webpage has a large stationary probability with many

links going towards it, is calculated.

Using the PageRank algorithm, we can construct a transition matrix T based on the transition probabilities defined by links from one page to another. The pages are the possible different states and the state vector is the PageRank vector with each element corresponding to the PR score of each page. Note that the PR of each page is dependent on the number of inbound and outbound links for each page u .

The PageRank also utilizes a damping factor and jumping. The goal of jumping is to connect all nodes or pages of the web to all other nodes, so if the random surfer finds themselves on a page that links to no other page they can arbitrarily jump to a random page. The damping factor, denoted by β , is the fraction of time the random web surfer spends clicking on links on the current page and $1 - \beta$ is the fraction of time the web surfer teleports or randomly jumps to other links in the network. The damping factor also has the effect of lowering the probability that the random surfer will continue to click links on each page.

The page rank for each page u in a collection of U pages of size N is

$$PR(u) = \frac{1 - \beta}{N} + \beta \cdot \sum_{v \in U, v \neq u} c(v) \frac{PR(v)}{L(v)} \quad (9)$$

where $c(v)$ is a coefficient of 1 if there is an outbound link to page u from page v and 0 if there is not an outbound link to page u from page v , $L(v)$ is the number of links page v contains, and N is the number of pages in the web.[24] [25] [26]

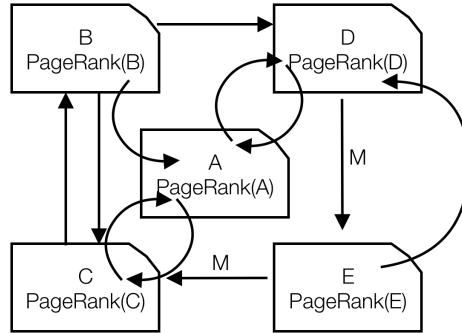


Figure 7: State diagram representing the PageRank algorithm with transitional probabilities of $M = PR(u)$. [36]

5.2 Covid 19

Moreover, Markov models can be used to predict the number of deaths and recovered patients. Provided with this information, specialists can assess whether a specific disease is an epidemic one.

One such research has been made in 2021 about COVID-19 by mathematicians Safaa K. Kadhem and Sadeq A. Kadhim [1]. In the health system, patients move from one state to another. Specifically, for COVID-19, individuals go through four states, two of which are transition states and the other two are absorbing states.

Absorbing states in a Markov chain are the entries of the transition matrix that equal to 1 when $i = j$ (rows=columns), which indicates that when entered that state, there is no possibility of leaving it.

The states that are described as transition states are: (a) susceptible, people who have been tested and (b) infectious, people who have the disease, got diagnosed, and are confirmed cases. The absorbing states are: (c) recovered, people who had the disease but are now healed, and (d) dead, people who died while having the disease.

With the provided data, a collection of all the individuals in each state was captured for 3 months in Iraq, and with the properties mentioned in Chapter 1 of the article, a transition matrix P was computed, which showed that only 5% of individuals are infected, compared to the susceptible 95%. Then, from the transition matrix P , the fundamental matrix of P was computed. The fundamental matrix W computes the expected time in any state and the time spent in the transient state j , starting from any transient state i . It is computed by taking the inverse of the identity matrix I minus the matrix Q , which reflects the transition probability from a transient state to another transient state (not absorbing states). $W = (I - Q)^{-1}$.

In order to obtain the life expectancy of the susceptible and infectious people, the rows of the matrix W were summed. This showed, approximately, the number of days a person is susceptible and infectious before moving into an absorbing state. Subsequently, another transition matrix Z was computed, in which the probabilities of transitioning from a transient state to an absorbing state were included. By multiplying the mean of individuals in susceptible and infected states by this matrix, a prediction of the number of dead and recovered patients was established.

The specific study [1] came to the conclusion that the percentage of deaths was gradually increasing, while the percentage of individuals who recovered was decreasing.

5.3 HIV

In addition, a Markov chain can be used to model the progression of AIDS among people, which can be helpful for the development of campaigns to reduce the infection rate.

A person that is carrying the HIV virus, can progress to having AIDS disease and

then die. Markov chain can be a useful tool, to predict that chance of progression. There are 4 states for every individual, S1: people who are vulnerable to the virus, S2: people who carry the HIV virus, S3: people who got diagnosed with AIDS, S4: people who have died from AIDS. The first three states, S1, S2, S3 are transient states and S4 is an absorbing state, as people who have died, cannot leave that state.

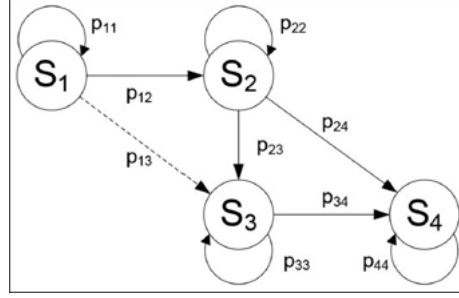


Figure 8: Markov chain diagram of transitioning states of patients [2]

Figure ‘8’ represents the states and the probabilities of transitioning from one state to another. (Note: the transition between states S1 and S3 cannot occur clinically, but it represents the individuals who never got the diagnosis of carrying the virus HIV and got immediately diagnosed with the disease).

Firstly, the data collected from the Centers of Disease Control and Prevention database between 2006-2009 is analyzed [2]. The data was given as rate r per 100000 people for each category, so the rates are computed by dividing the number of cases reported by the population for each year and multiplying them by 100000. These rates are then converted to transition probabilities and the transition probability matrix is computed.

By this transition matrix, findings can be made on how many years a patient can go carrying the HIV virus without having any symptoms, as on average most patients do not experience symptoms until getting AIDS, so the time for transition from state S2 to S3 can indicate the time needed to experience HIV symptoms. It has been found, according to research conducted by Clement Twumasi, Louis Asiedu, and Ezekiel N. N. Nortey [4], that approximately 14.29 years can pass in order for an individual to have some sort of symptom related to HIV and not the common flu. Moreover, the transition from state S3 to S4 can be an indicator of the approximation of years that a person can have AIDS before dying, which has been found to be 20 years, [4]. (the research was done in 2016, the medical field has grown since then and new treatments have been established, we just want to illustrate how Markov chains can be used in real-life). Lastly, by the use of Markov chains, it was shown that the number of HIV/AIDS diagnoses and the number of related deaths each year, between 2006-2009, was quite consistent, on the other hand, the number of living individuals with the virus or the disease was growing [2].

5.4 Diabetes

Markov chains can also be used to help people affected by diabetes. By using its method, not only the levels of blood sugar can be predicted, in order for the patients to take the necessary precautions, but the progression of a common complication of diabetics, called diabetes retinopathy (DR) can be modeled.

Diabetes is a disorder where the levels of blood sugar become higher, due to the lack of insulin or the malfunction of insulin receptors. It has become one of the largest causes of death and people suffering from it are increasing. It can cause various complications, if not treated correctly. By using the Markov model, the anticipation of blood sugar levels in individuals with diabetes is enabled, thereby facilitating the implementation of appropriate measures. There are 3 states of blood sugar, low, normal, and high, s_1 , s_2 , s_3 respectively and all states are transient, as blood sugar levels can transition at each state. By measuring the probability of transition from each state, a transition probability matrix can be formed. From this matrix, vital information can be derived, as an individual can ascertain the likelihood of experiencing elevated blood sugar levels within a specific time frame, thereby enabling them to make informed decisions regarding their dietary choices in order to avoid particular foods, or assert the time needed to get low blood level sugar, ensuring the proper administration of their medication. [17]

Diabetics can experience a number of complications, one of them being DR which is one of the most common causes of visual loss. There are 5 stages of DR, which will be the states in the Markov model: S_1 =mild non-proliferative diabetic retinopathy (NPDR), S_2 =moderate NPDR, S_3 =Severe NPDR, S_4 = proliferative diabetic retinopathy (PDR) and S_5 = blindness. The first 4 states are transient states, whereas the last state is an absorbing state.

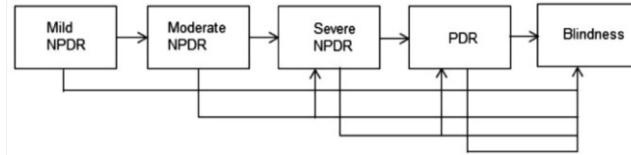


Figure 9: states of progression [5]

To begin, with data of 153 patients, collected from the “Eye camp” in 2010 [5], a transition matrix M is computed, with all the probabilities of moving from one state to another in the cycle of diabetic retinopathy in one year. Analyzing it, gave the following findings (for that year):

- 80% of patients that were in state 1, remained in that state, 12% moved to state 2 and the rest 8% to state 3
- 44% of patients in state 2, remained in the state, 30% moved to state 3 and 26% to state 4

- 46% of patients in state 3, stayed in state 3, 50% moved to state 4 and 4% became blind (state 5)
- 88% of patients in state 4, stayed in that state and the rest 12% transitioned to state 5

In order to find the expected probability of each patient moving to another state in the span of more years, a multiplication of each successive power of the initial transition matrix with the initial state vector has to be calculated $P(n)=P(n-1)*P$, so the initial state vector is multiplied by the first power of the matrix, that result is then multiplied by the second power and so on. For instance, the state at year 3 will be equal to $P(0)*P^2$. Following, a stationary Markov chain matrix P is computed, which is a partitioned matrix of the matrices $M, R, 0, P1$.

$$P = \begin{pmatrix} M & R \\ 0 & P1 \end{pmatrix}$$

where M is the transition matrix computed earlier, R is the transition matrix for moving from the transition states to the recurrent states (if we leave those states, we return to them with probability equal to 1), 0 is the zero matrix which represents the zero probabilities of moving from recurrent states to transitioning states and $P1$ represents the transition matrix of transitioning between recurrent states. The amount of time spent in a transient state, before moving to a recurrent state, can be found by summing the i^{th} row of $(I - M)^{-1}$. After P is computed, by the information gathered [5], a patient in mild NPDR (state 1) is likely to stay there for the next 5 years, in moderate NPDR (state 2) is most probably going to stay there for 1.07 years, in severe NPDR (state 3) for 1.33 years and in PDR (state 4) for 7.88 years. Hence, when patients get the specific complication of diabetes, it takes approximately 15.28 years for them to get completely blind.

In light of this information, Markov chains have been proven to be hidden in many aspects of the real-life world and can be a helpful tool for solving or controlling various problems.

6 Population Growth

6.1 Introduction on Population Growth

As we already observed in the article, Markov Chains have various examples where they can be applied. Population growth is one of those applications in which a Markov process can be observed for different reasons. Now, we will observe how Markov Chains are adapted to the birth-death problem. Population growth can also be modeled as a birth-death process in a random environment where the environmental changes are influenced by the population size (say, in the case of an overpopulation). Generally, Birth-death processes are fundamental stochastic models in applied probability, particularly in queueing (Queueing theory is the model that is being constructed which allows us to predict the behavior of a system only by a single queue or a series

of queues). Also, most of our focus goes to models where the transitions of birth-death processes are affected by the environment. Markov chains for those stochastic models allow for verification of the existence of stationary distribution via analysis of various conditions. Using Markov Chains for those stochastic models can be a check for being in a steady state or not.[14, 15, 16]

6.2 Helpful Information

For starters: [16] Consider a birth-death process $N = \{N(t), t \geq 0\}$ on $N = \{0, 1, 2, \dots\}$ with birth rates $\lambda_n(z)$ (for the jump $n \rightarrow n + 1, n \geq 0$) and death rates $\mu_n(z)$ (for the jump $n \rightarrow n - 1, n \geq 1$) depending on an environment variable z taking values in an environment space Z . The environmental variable, in turn, evolves as a continuous-time Markov process $Z = \{Z(t), t \geq 0\}$, with transition functions depending on the current state $N(t)$ of the birth-death process (basically we deal with a group of birth-death and environmental processes depending on the parameters $z \in Z$ and $n \in \mathbb{N}$). A combined Markov process $(N, Z) = \{(N(t), Z(t)), t \geq 0\}$, with states (n, z) , is called a birth-death process in an interactive random environment.[12]

The generator $\mathcal{L}$ of the combined process (N, Z) is given by:

$$\mathcal{L}f(n, z) = M_z f(n, z) + r_n(z)^{-1} A_n f(n, z).$$

Additionally, this article will concentrate on a model of the environment space Z where is at most countable, and the environment process is a continuous-time Markov chain with a generating matrix A_n depending on n and the state of process $N(t)$; we call this model a *jump environment*.

For that model, we will consider:

- (a) the existence and uniqueness of a stationary distribution, that is, a probability distribution π on the product space $X = N \times Z$ such that if $(N(0), Z(0)) \sim \pi$, then $(N(t), Z(t)) \sim \pi$ for all $t \geq 0$, and an explicit form of π ;
- (b) convergence $(N(t), Z(t)) \rightarrow \pi$ as $t \rightarrow \infty$ in the total variation distance, and the rate of this convergence (exponential or polynomial).

In the article the definitions of finite invariant measure and unique invariant distribution are being used. In order to understand those terms, we need to look at dynamics systems theory.

Firstly, a measure is a mathematical concept that assigns a value to subsets of a set, often used to describe the probability or density of points or sets of points in the system. Measures can be used to quantify various properties of the system, such as its distribution, behavior, or statistical properties.

An invariant measure is a measure that remains constant or unchanged when the system evolves over time according to its dynamics. In other words, if a measure is invariant, its value or distribution does not change as the system undergoes its evolution. A finite invariant measure specifically refers to an invariant measure that

has a finite total measure. This means that the total measure assigned to the entire space or set under consideration is finite.

Finite invariant measures play an important role in the study of dynamical systems, as they provide insights into the long-term behavior and properties of the system. They can help identify and characterize stationary or equilibrium states and provide information about the statistical properties of trajectories or orbits within the system. The term unique invariant distribution, used later on, refers to a specific property of a stochastic process or dynamical system. It indicates that there exists only one probability distribution that remains invariant or unchanged under the evolution of the system.

Also, we use some queueing theory for the examples below. Firstly, M/M/∞ queue appears in the first example. An M/M/∞ queue is a type of queueing system that follows a specific probability model. The “M” stands for “Markovia”, indicating that the arrivals and service times in the queue are exponentially distributed and independent of each other. The second “M” represents the same property for the service times. The “∞” indicates that the system has an infinite number of servers available. Some properties of an M/M/∞ are memorylessness (the current event does not depend on the system’s past behaviour) and its service times are exponentially distributed. Finally, the definition offered load appears. That refers to the rate at which the elements the example is dealing with are being introduced or “offered” to a queueing system.

Observing [11], we see the evolution of a new mutant population. These mutant populations evolve under the laws of a continuous time Markov chain that has its unique transition probabilities. An important assumption is that all populations evolve under the same Markov chain. This comes from the fact that each Markov chain has a unique stationary probability matrix and this could lead to potentially false results. In addition, each population is independent of one another.

[11] comes to the result that a birth and death process is a stationary Markov process with a transition probability matrix:

$$P_{ij} = \begin{cases} \lambda_i t + o(t) & \text{if } j = i + 1 \\ \mu_i t + o(t) & \text{if } j = i - 1 \\ 1 - (\lambda_i + \mu_i)t + o(t) & \text{if } j = i \end{cases}$$

6.3 Notations and Assumptions

In order to understand all the examples that are going to appear afterwards, we need to state some assumptions (which we will not prove since it is out of our scope) as well as notations and some definitions which are being used onwards.

Let Z be a finite or countable state space for the environment. We define a two-component Markov process (N, Z) taking values in the countable state space $N \times Z$, with the following generator matrix $R = (R[(n, z), (n', z')])$: [16]

$$R[(n, z), (n + 1, z)] = \lambda_n(z), \quad R[(n, z), (n - 1, z)] = \mu_n(z),$$

$$R[(n, z), (n, z)] = r_n(z)^{-1} \tau_n(z, z), \quad R[(n, z), (n', z')] = 0, \quad n = n', \quad z = z',$$

where for each $z \in Z$,

$$r_n(z) = \prod_{k=1}^n \frac{\lambda_{k-1}(z)}{\mu_k(z)} \quad \text{for } n \geq 1, \quad \text{and} \quad r_0(z) \equiv 1$$

and $T_n = (\tau_n(z, z'))_{z, z' \in Z}$ is the generator for an irreducible continuous-time Markov chain on Z . Here $N = \{N(t) : t \geq 0\}$ represents the dynamics of the birth-death process, taking values in N , and $Z = \{Z(t) : t \geq 0\}$ indicates the evolution of the environment. It can be easily checked that for each $z \in Z$, we have the detailed balance equations for the birth-death process $N(t)$: [16]

$$\begin{aligned} \kappa_n(z)(\lambda_n(z) + \mu_n(z)) &= \kappa_{n-1}(z)\lambda_{n-1}(z) + \kappa_{n+1}(z)\mu_{n+1}(z), \quad n \geq 1, \\ \kappa_0(z)\lambda_0(z) &= \kappa_1(z)\mu_1(z), \end{aligned}$$

where

$$\begin{aligned} \kappa_n(z) &= \frac{\kappa_0(z)}{\sum_{k=1}^n \frac{\lambda_{k-1}(z)}{\mu_k(z)}}, \quad n \geq 1, \\ \kappa_0(z) &= \left(1 + \sum_{j=1}^{\infty} \prod_{k=1}^j \frac{\lambda_{k-1}(z)}{\mu_k(z)} \right)^{-1}. \end{aligned}$$

Last assumption: [16]

$$\sum_{(n, z)} r_n(z) v(z) = \sum_{(n, z)} \frac{\rho(z)^n}{n!} v(z) = \sum_{(n, z)} \left(\sum_{k=1}^n \frac{\lambda_{k-1}(z)}{\mu_k(z)} \right) v(z) < \infty$$

6.4 Examples

[16] There are some examples of the usage of Markov Chains in Population Growth:

Example 1

For an M/M/ ∞ queue, $\lambda_n(z) = \lambda(z)$ and $\mu_n(z) = n\mu(z)$ for all n and $z \in Z$.

Hence,

$r_n(z) = \frac{\rho(z)^n}{n!}$, where $\rho(z) := \frac{\lambda(z)}{\mu(z)} \in (0, \infty)$ is the offered load. The sum of all these $r_n(z)$ is

$$\sum_{n=0}^{\infty} r_n(z) = e^{\rho(z)}$$

Thus, after normalizing, we get

$$\kappa_0(z) = e^{-\rho(z)} \quad \text{and} \quad \kappa_n(z) = e^{-\rho(z)} \frac{\rho(z)^n}{n!}, n \geq 1, z \in Z.$$

From assumption, this takes the following form:

$$\Xi = \sum_{(n,z)} \frac{\rho(z)^n}{n!} v(z) = \sum_z e^{\rho(z)} v(z) < \infty$$

That concludes the example since:

$$\eta(n, z) = \frac{\rho(z)^n v(z)}{n!} \text{ yields a finite invariant measure for } (N, Z), \text{ and}$$

$$\pi(n, z) = \frac{\eta(n, z)}{\Xi} \text{ is a unique invariant probability distribution.}$$

Example 2

[16] This example is a bit more complicated since it involves a linear population growth model with immigration:

For each $z \in Z$, $\lambda_n(z) = n\lambda(z) + \theta(z)$ for $n \geq 0$ and $\mu_n(z) = n\mu(z)$ for $n \geq 1$. In this model, each individual in the population gives birth at a rate $\lambda(z) > 0$; in addition, there is an exponential rate of growth of the population $\theta(z) > 0$ due to an external source (immigration). The death rate is given by $n\mu(z)$. Hence,

$$r_n(z) = \frac{\prod_{k=1}^n ((k-1)\lambda(z) + \theta(z))}{n!\mu(z)^n} = \frac{\rho(z)^n}{n!} \prod_{k=1}^n \left(\frac{k-1 + \theta(z)}{\lambda(z)} \right)$$

for $n \geq 0, z \in Z$.

where $\rho(z) = \frac{\lambda(z)}{\mu(z)} \in (0, \infty)$. We also have

$$\kappa_0(z) = \left(1 + \sum_{j=1}^{\infty} \frac{\rho(z)^j}{j!} \prod_{j=1}^{\infty} \left(\frac{(k-1) + \theta(z)/\lambda(z)}{j} \right) \right)^{-1},$$

$$\kappa_n(z) = \kappa_0(z) \frac{\rho(z)^n}{n!} \prod_{n=1}^{\infty} \left(\frac{(k-1) + \theta(z)/\lambda(z)}{j} \right), \quad n \geq 1.$$

By assumption, in this case, is

$$\sum_{(n,z)} r_n(z) v(z) = \sum_{(n,z)} \frac{\rho(z)^n}{n!} \prod_{n=1}^{\infty} \left(\frac{(k-1) + \theta(z)/\lambda(z)}{j} \right) v(z) < \infty.$$

So, a finite invariant measure on $N \times Z$ has the desired form:

$$\eta(n, z) = r_n(z)v(z) = \frac{\rho(z)^n}{n!} \prod_{j=1}^{\infty} \left(\frac{(k-1) + \theta(z)/\lambda(z)}{j} \right) v(z), \quad \text{for } n \geq 0, z \in Z.$$

All in all, we see that Markov chains play a vital role in the study of population growth.

7 Conclusion

A Markov chain, defined in terms of a set of states and a transition model from one state to another, is a crucial mathematical model depicting the independence of a sequence of events from any past states other than the current state at a given time. The model proves very useful as it can be used to predict various states, which can affect different aspects of life. By studying the theory behind a Markov chain we observed that many different fields incorporate this model in various applications, some of which do so in combination with the notion of relaxation time, which defines the time period needed for a system to return to its initial states after undergoing a disturbance during a transition from one state in a Markov chain to the next.

The probability of a system remaining in each state after a long period of time (so the long-term probability) is the steady-state behaviour of a Markov chain. As discussed in the article, not all Markov chains reach a steady state, meaning that the “current” behaviour of the system does not continue to the future for those systems. Having delved into the mathematics of Markov chains and examined a variety of Markov models we come to the conclusion that in order for a Markov chain to reach an equilibrium, it has to be aperiodic and irreducible.

By studying various applications, we were able to determine a link between Markov chains and relaxation time, however, we encountered limitations on more in-depth information regarding the relationship and association between the two concepts. This suggests that several promising avenues for further research can be explored, building upon the insights gained from this study, to enhance our understanding of this topic as well as its value in our daily lives.

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